



HMC Project HARMONise

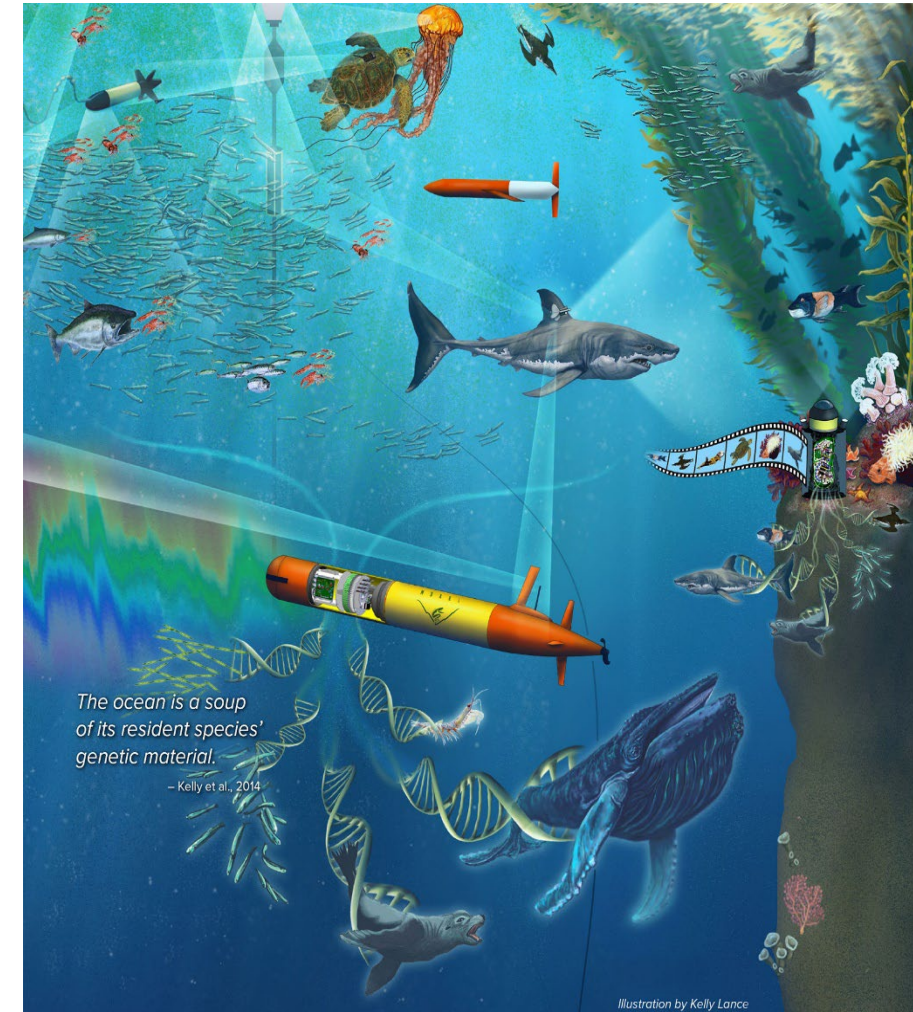


**Enhancing interoperability of
marine biomolecular (meta)data
across Helmholtz Centres**



Till Bayer, Christina Bienhold, Lars Harms, Stefan Neuhaus, Roland Koppe, Isabell Siebert

- Project routed in Research Area Earth & Environment
 - All ocean life is built from „biomolecules“ such as **DNA** and **RNA**
 - Biomolecular data provides insights into the **structure and function of marine communities in space and time**
 - Biomolecular (meta)data has **great internal diversity and complexity**
 - To date, biomolecular (meta)data management is not well integrated across Helmholtz Centres
 - A lack of interoperability prevents efficient exploitation of available datasets, as well as FAIR data exchange internally and externally
- Resolving it will result in an added value for researchers and Helmholtz



Chavez et al. 2021

<https://doi.org/10.5670/oceanog.2021.218>

sequencing data (e.g. fastq file)

```
@M00493:291:000000000-B55PM:1:1101:18919:1817 1:N:0:CCTAAGAC+AAGGCTAT
TGTCTACGGGAGGCTGCAGTGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCAACGCCGCGTGTGTGA
+
CCC@CE,CFF>7:@FG?FFGEFGGG,@E,CEFFCD<,C<,,;6:B7+6CF,,;<C,,9CEF7:@+77@+,,,9,
@M00493:291:000000000-B55PM:1:1101:11635:1892 1:N:0:CCTAAGAC+AAGGCTAT
CCTACGGGAGGCTGCAGTCGAGAATCTTCGGCAATGGACGAAAGTCTGACCGAGCGACGCCGCGTGC GGGATGA
+
CCCCCGDGGGGGGGGGGGDFFCGFC,EEGGFGGGGGGGG9F@F>EGGCFGFE@FEFCGGGDEEGGGGG7=>+>CF<
@M00493:291:000000000-B55PM:1:1101:15699:1975 1:N:0:CCTAAGAC+AAGGCTAT
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```

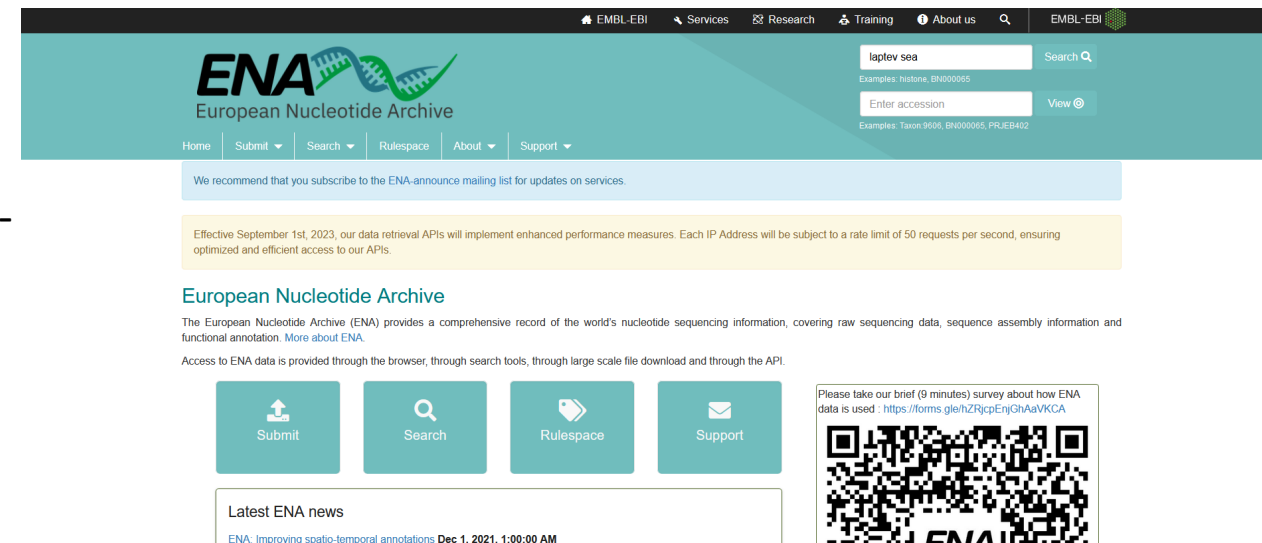
What **When** **Where** **Who** **How** **Depth** **Elevation** **Latitude** **Longitude** **Particle** **Water** **Soil** **Sediment** **Air** **Date** **Time** **Gear** **DNA extraction** **library strategy** **Filter size** **Sequencing platform** **Christina Bienhold** **Swantje Ziemann**

International framework

ENA – European Nucleotide Archive



- **International Nucleotide Sequence Database Collaboration** (INSDC: ENA, National Center for Biotechnology Information - NCBI and the DNA Data Bank of Japan - DDBJ)
- INSDC databases provide a **comprehensive record of the world's nucleotide sequencing information**, covering raw sequencing data, sequence assembly information and functional annotation



MlxS – Minimum Information about any (x) sequence

Yilmaz, P., Kottmann, R., Field, D., Knight, R., Cole, J.R., Amaral-Zettler, L., et al. (2011). *Nature Biotechnology* 29(5), 415-420. doi: 10.1038/nbt.1823.

- INSDC databases adopt **MlxS as a metadata standard**
- **Standard** developed by the Genomics Standards Consortium (GSC) for reporting sequence data
- **Checklists** for different sets of data, e.g. genomes, meta-genomes, marker genes etc.
- **Environmental packages** with checklists for specific environments (e.g. sediment, water)
- **Environment Ontology (EnvO)**: community ontology for the concise, controlled description of environments



	genomes	meta-genomes	marker genes	
Specification projects	MIGS	MIMS	MIMARKS	New checklists
Checklists	EU BA PL VI ORG	metagenomes	survey specimen	e.g., pan-genomes
Shared descriptors	collection date, environmental package, environment (biome), environment (feature), environment (material), geographic location (country and/or sea, region), geographic location (latitude and longitude), investigation type, project name, sequencing method, submitted to INSDC			
Checklist-specific descriptors	assembly, estimated size, finishing strategy, isolation and growth condition, number of replicons, ploidy, propagation, reference for biomaterial		target gene	
Applicable environmental packages (measurements and observations)	Air Host-associated Human-associated Human-oral Human-gut Human-skin Human-vaginal Microbial mat/biofilm Miscellaneous natural or artificial environment Plant-associated Sediment Soil Wastewater/sludge Water			

dark data

[...] data that has never been published or otherwise made available to the rest of the scientific community.

B. P. Heidorn Libr. Trends 57, 280–299; 2008

dark (dirty) metadata

data submitted to sequence databases often do not comply with metadata standards

→ Interoperability and reusability of nucleotide sequencing data is still limited (e.g. Hassenrück et al. 2021 bioRxiv)

Adapted from: Ivaylo Kostadinov, GFBio e.V.

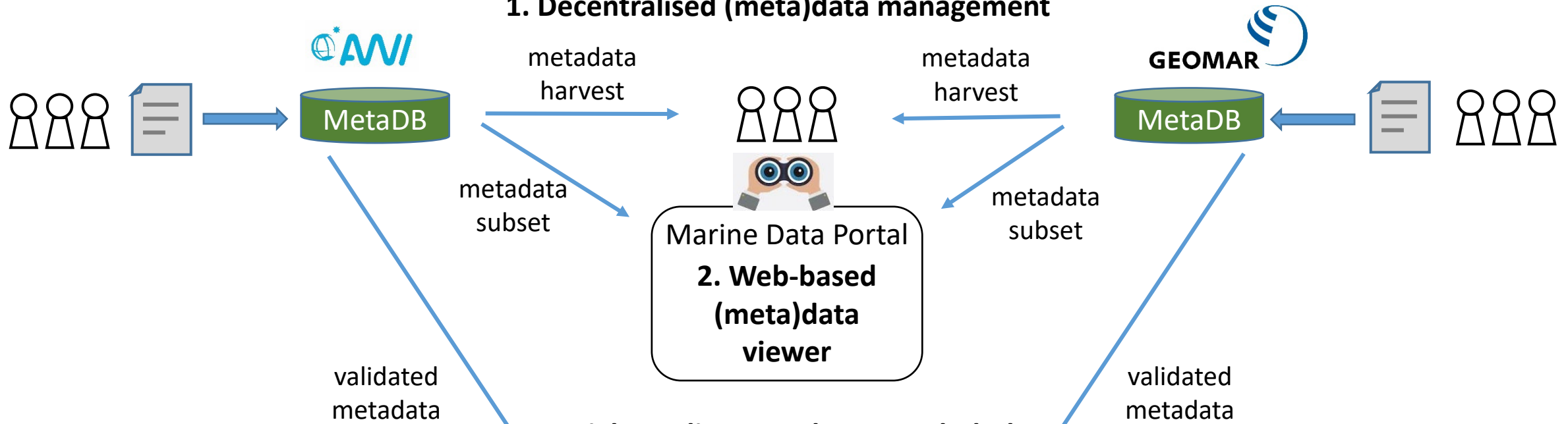
The project aims to develop sustainable local-to-global workflows that support:

- 1. High-quality curation and management of marine biomolecular metadata**
- 2. FAIR exchange and publication of (meta)data within HMC systems and globally**
- 3. Interlinking biomolecular (meta)data to other Helmholtz digital holdings and services**

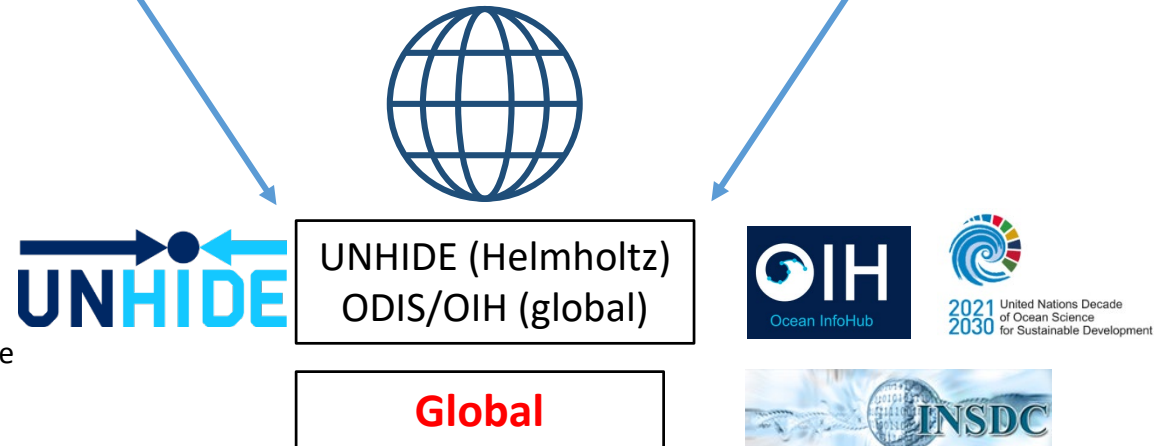
Project approach

Local

1. Decentralised (meta)data management



3. High-quality metadata to Helmholtz and global digital ecosystem



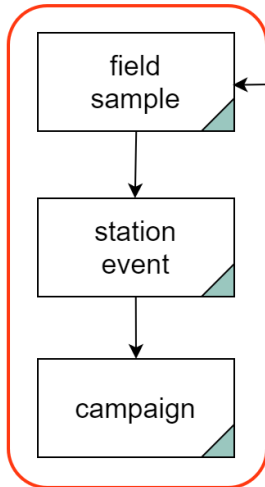
HELMHOLTZ
METADATA COLLABORATION



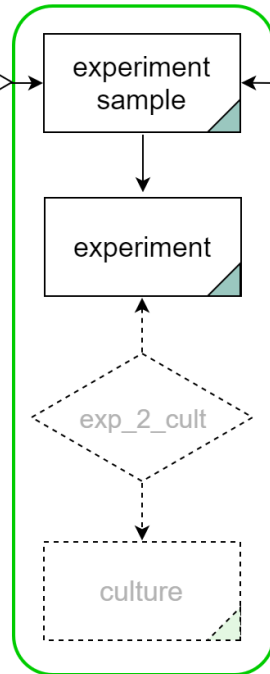
PANGAEA.



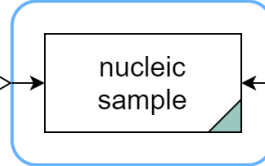
field-related
metadata



experiment-related metadata



metadata associated
with nucleic acid
extraction

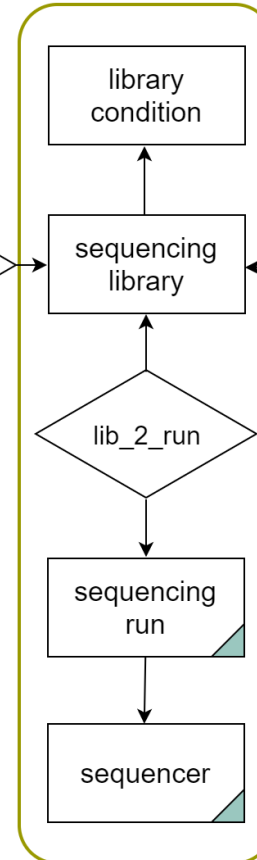


connections to this
entity is indicated
by corner color
coding

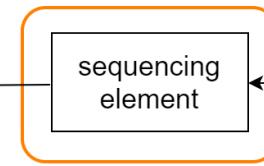


ORCID

metadata associated
with the sequencing
process



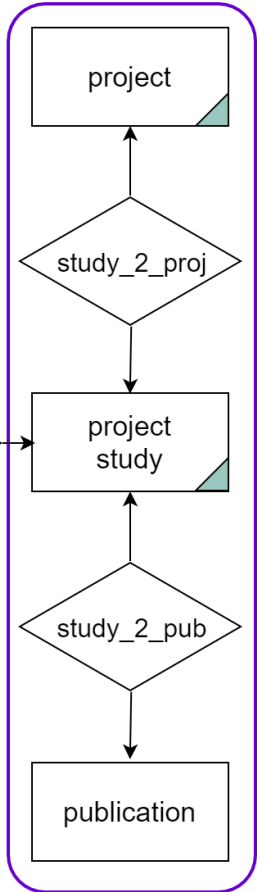
metadata associated
with the sequence
raw data files



Alignment with MLxS standard of the Genomic Standards Consortium



project-related
metadata



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Welcome

19
sequencing elements in database

11
campaigns in database

55
samples in database



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Data Import

 GFBio_template Freigebe

Start Einfügen Zeichnen Seitenlayout Formeln Daten Überprüfen Ansicht

B1

	A	B	C	D	E	F	G	H	I
1	sample_title	forward_read_file_name	forward_read_file_checksum	reverse_read_file_name	reverse_read_file_checksum	checksum_method	investigation type	taxon_id	sample_id
2									
3									
4									
5									
6									
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8									
9									
10									
11									
12									
13									
14									
15									

 Tabelle2 +

Zelle B1: kommentiert von Lars Harms

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HARMONise Exchange Hub

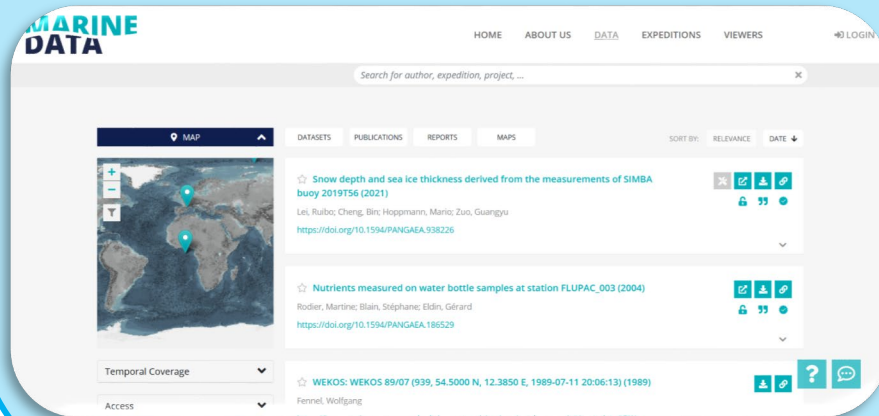
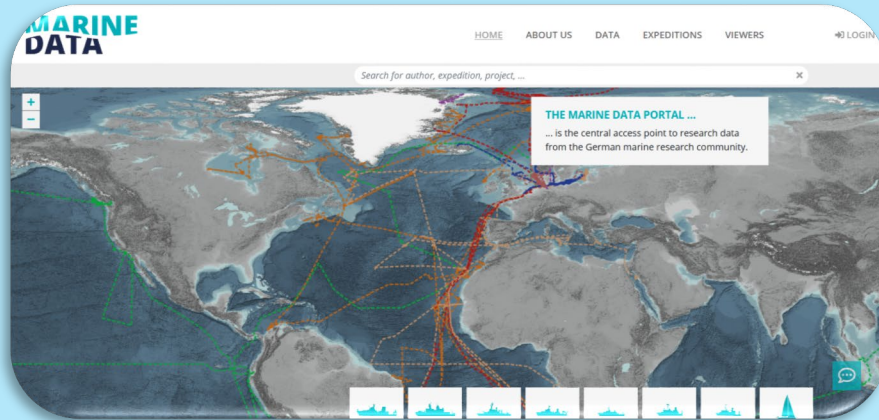
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- Import metadata compliant with the international MlXS standard (Genomic Standards Consortium)
- Harmonise metadata entries with existing Helmholtz repositories (e.g. PANGAEA World Data Center, sample management systems)
- Enable standardized exchange of core metadata
- Promote global interoperability by exporting metadata aligned with international standards and data exchange formats (e.g. UNESCO ODIS-Arch specifications)

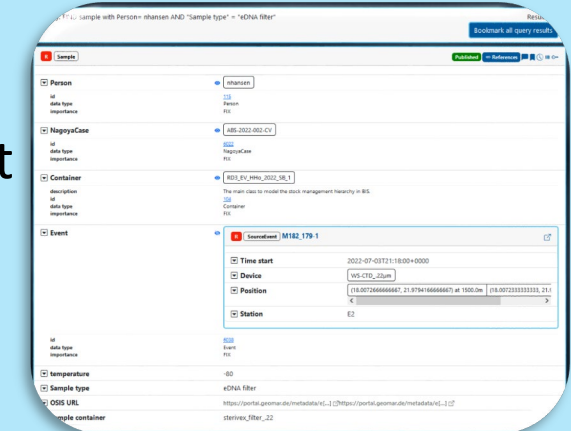


Links to national and international initiatives

Marine-data portal



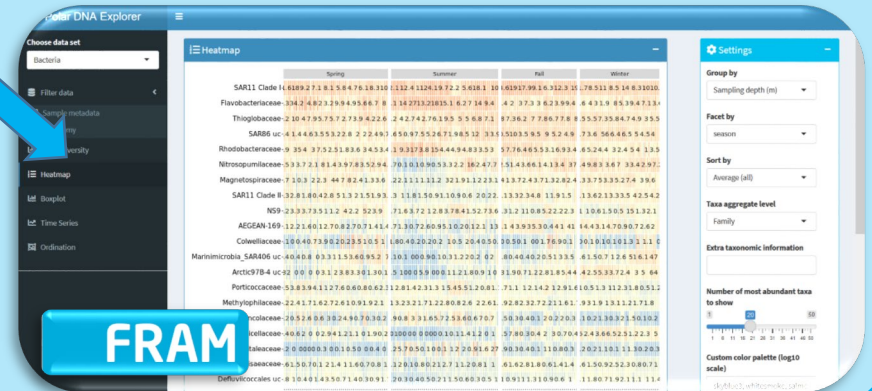
Sample management systems



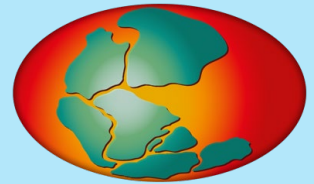
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Data exploration tools



Links to national and international initiatives



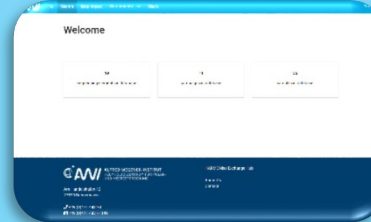
PANGAEA.

PANGAEA
World Data Center

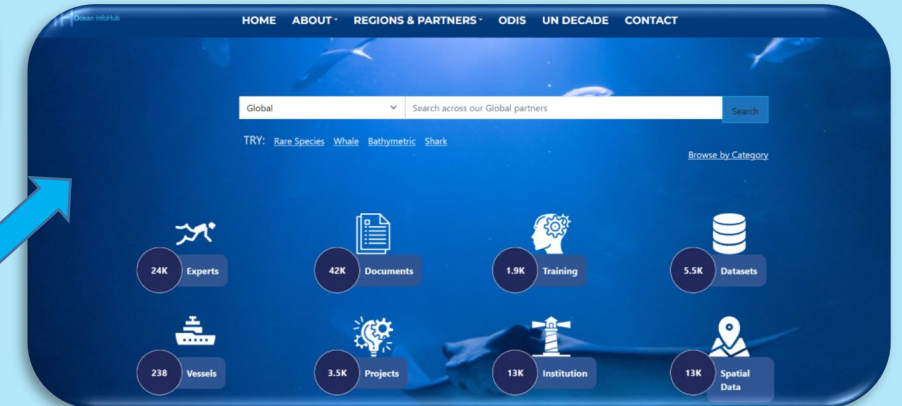
Helmholtz knowledge graph
UNHIDE - unified Helmholtz
Information and Data Exchange



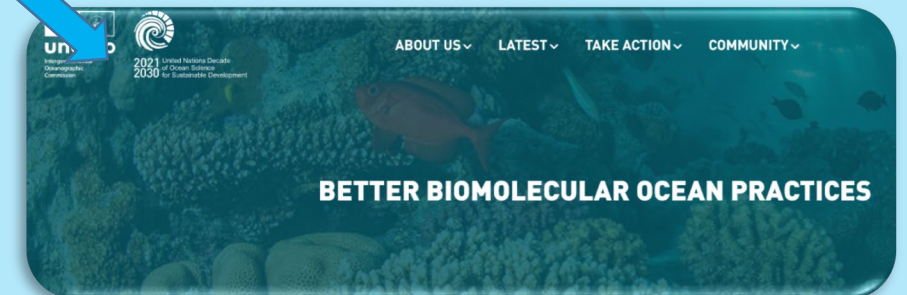
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Ocean Info Hub



UN Decade Project BeBOP
Better Biomolecular Ocean Practices



OBCON
OCEAN BIOMOLECULAR
OBSERVING NETWORK

- Support the **development of digital cultures** in the Centers by providing a framework for high-quality and standards-compliant management of marine biomolecular metadata
- Enhance **biomolecular (meta)data exchange**, interfacing with and enhancing the Helmholtz digital ecosystem
- Enhance **Helmholtz-to-global interoperability** through the use of international standards, associated ontology content, and data exchange formats

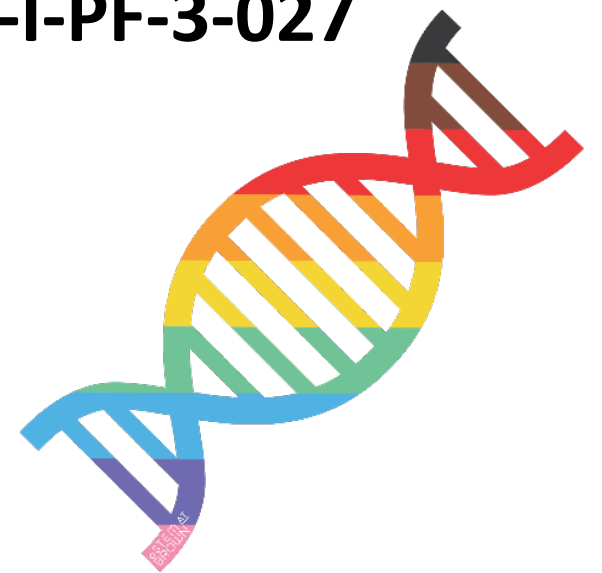


Thank you



**Helmholtz Initiative and Networking Fund ZT-I-PF-3-027
HMC Hub Earth & Environment**

You for your attention



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